

1. (a) As an example, for the empirical application of parents' education and baby's birth-weight, the following joint null hypothesis imposes two restrictions ($q = 2$):

$$H_0 : \beta_1 + \beta_2 = 0.1, \text{ and } \beta_3 = 0, \quad (1)$$

which means the combined effect of parents' education is equal to 0.1 (pound), and the effect of smoking is zero. Its alternative hypothesis is

$$H_1 : \text{at least one restriction in } H_0 \text{ does not hold.} \quad (2)$$

This alternative hypothesis is equivalent to (when $q = 2$ there are 3 possible cases):

$$H_1 : \beta_1 + \beta_2 \neq 0.1, \text{ or } \beta_3 \neq 0, \text{ or neither restriction holds.} \quad (3)$$

[In the above example, there are 3 possible cases under H_1 . In general, if there are q restrictions in H_0 , there will be $2^q - 1$ different cases under H_1 . For example, if $q = 1$, then $2^1 - 1 = 1$; if $q = 2$, then $2^2 - 1 = 3$; if $q = 3$, then $2^3 - 1 = 7$; if $q = 4$, then $2^4 - 1 = 15$, etc. This formula is just for your reference, so no need to recite it.]

Also refer to the lecture slides for other empirical examples.

- (b) For example, consider the following regression model:

$$sleep_i = \beta_0 + \beta_1 totalwork_i + \beta_2 educ_i + \beta_3 age_i + u_i, \quad i = 1, \dots, n. \quad (4)$$

This is the unrestricted regression model (full regression model), and we compute R_u^2 from this model. Suppose we want to test the joint null hypothesis $H_0 : \beta_2 = \beta_3 = 0$, so that $q = 2$ in this case. Then, the restricted regression model is the model with H_0 imposed:

$$sleep_i = \beta_0 + \beta_1 totalwork_i + u_i, \quad i = 1, \dots, n, \quad (5)$$

from which we compute R_r^2 .

- (c) The formula of the homoskedasticity-only F -statistic is as follows:

$$F = \frac{(R_u^2 - R_r^2)/q}{(1 - R_u^2)/(n - k - 1)}, \quad (6)$$

where

- R_u^2 : the R^2 from the unrestricted regression (the full regression model),
- R_r^2 : the R^2 from the restricted regression under the restrictions in H_0 ,

- q is the number of restrictions in H_0 , k is the number of regressors in the unrestricted regression (intercept not included in k), and n is the number of observations.

The properties of the homoskedasticity-only F -statistic are as follows:

By construction, $F \geq 0$, which is because of the fact that R^2 increases with the number of regressors, so that R_u^2 cannot be smaller than R_r^2 . The homoskedasticity-only F -statistic is comparing the fittings of the two regressions; if the “unrestricted” model fits the data sufficiently better than the “restricted” model, this can be understood as an evidence against the joint null hypothesis H_0 . In this case, we will reject the “restricted” model (that is, we will reject H_0).

- (d) Given that $\sum_{i=1}^n \hat{u}_i = 0$, we have

$$\sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{1i} - \dots - \hat{\beta}_k X_{ki}) = 0. \quad (7)$$

This implies

$$\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{1i} - \dots - \hat{\beta}_k X_{ki}) = 0, \quad (8)$$

which is equivalent to

$$\bar{Y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{X}_1 - \dots - \hat{\beta}_k \bar{X}_k = 0, \quad (9)$$

where $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$, $\bar{X}_1 = \frac{1}{n} \sum_{i=1}^n X_{1i}$, ..., and $\bar{X}_k = \frac{1}{n} \sum_{i=1}^n X_{ki}$. If $\hat{\beta}_1 = \dots = \hat{\beta}_k = 0$, then $\bar{Y} = \hat{\beta}_0$.

On the other hand, by the definition of the OLS predicted value $\hat{Y}_i$,

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \dots + \hat{\beta}_k X_{ki}. \quad (10)$$

Therefore, if $\hat{\beta}_1 = \dots = \hat{\beta}_k = 0$, then $\hat{Y}_i = \hat{\beta}_0$.

Finally, given the definition of ESS , in this case we have

$$ESS = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 = \sum_{i=1}^n (\hat{\beta}_0 - \bar{\beta}_0)^2 = 0, \quad (11)$$

which implies that $R^2 = 0$.

2. We use the OVB formula to discuss the likely bias in the OLS estimator of β_1 . Because the error term contains workers' innate ability, we can specify the true model as:

$$wage_i = \beta_0 + \beta_1 train_i + \beta_2 educ_i + \beta_3 experience_i + \beta_4 ability_i + e_i, \quad i = 1, \dots, n,$$

where “*ability*” is the omitted variable. By the meaning of multiple regression, β_4 is the effect of workers' innate ability (cognitive, physical, etc.) on wage (after controlling for the

other factors “*train*”, “*educ*”, and “*experience*”). β_4 is positive because we expect that on average the wage increases when a worker’s innate ability is relatively high.

Now we need to decide the sign of $Cov(train_i, ability_i)$.

(1) Suppose that the employer decided to send the workers with relatively bad performance to training, then less able workers have a greater chance of being selected for the training program, so that $Cov(train_i, ability_i) < 0$ and by the OVB formula, we have

$$OVB = \beta_4 \cdot \frac{Cov(train_i, ability_i)}{Var(train_i)} < 0.$$

That is, a downward bias in $\hat{\beta}_1$ in this scenario.

(2) Suppose that the employer decided to send the workers with relatively good performance to training, then more able workers have a greater chance of being selected for the training program. Therefore, in this case, $Cov(train_i, ability_i) > 0$ and $OVB > 0$ (upward bias).

(3) Suppose that the employer decided to randomly assign the workers to training, then we have $Cov(train_i, ability_i) = 0$ (because $train_i$ and $ability_i$ are independent in this case) and $OVB = 0$ (no bias).

3. (a) $\widehat{\Delta Price} = -56.9$ (\$1000).
- (b) $\widehat{\Delta Price} = 26.9 + 0.239 \cdot 80 = 46.02$ (\$1000).
- (c) For testing $H_0 : \beta_1 = 0$, we have

$$|t| = \left| \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)} \right| = \left| \frac{0.567}{1.23} \right| = 0.461 < 1.96. \quad (12)$$

Therefore, the coefficient on BDR is not statistically significantly different from zero at the 5% level. The 95% confidence interval for the coefficient on BDR is:

$$\begin{aligned} & [\hat{\beta}_1 - 1.96 \cdot SE(\hat{\beta}_1), \hat{\beta}_1 + 1.96 \cdot SE(\hat{\beta}_1)] \\ &= [0.567 - 1.96 \cdot 1.23, 0.567 + 1.96 \cdot 1.23] \\ &= [-1.8438, 2.9778]. \end{aligned} \quad (13)$$

- (d) For the multiple regression model

$$Price_i = \beta_0 + \beta_1 BDR_i + \beta_2 Bath_i + \beta_3 Hsize_i + \beta_4 Lsize_i + \beta_5 Age_i + \beta_6 Poor_i + u_i, \quad (14)$$

where $i = 1, \dots, n$, we would like to test the joint null hypothesis

$$H_0 : \beta_1 = 0, \text{ and } \beta_5 = 0. \quad (15)$$

The alternative hypothesis is

$$H_1 : \beta_1 \neq 0, \text{ or } \beta_5 \neq 0, \text{ or neither restriction holds.} \quad (16)$$

The unrestricted regression model is defined in (17). The restricted regression model can be written as

$$Price_i = \beta_0 + \beta_2 Bath_i + \beta_3 Hsize_i + \beta_4 Lsize_i + \beta_6 Poor_i + u_i, \quad i = 1, \dots, n. \quad (17)$$

4. We are interested in the following multiple regression model:

$$wage_i = \beta_0 + \beta_1 compwork_i + \beta_2 comphome_i + \beta_3 compwork_i \cdot comphome_i + u_i, \quad i = 1, \dots, n. \quad (18)$$

To simplify notation, let us denote $wage_{[compwork=0, comphome=0]} = E[wage_i | compwork_i = 0, comphome_i = 0]$.

$wage_{[compwork=1, comphome=0]}$, $wage_{[compwork=0, comphome=1]}$, and $wage_{[compwork=1, comphome=1]}$ are defined in a similar way.

Then, given that OLS Assumption 1 holds, we have

$$wage_{[compwork=0, comphome=0]} = \beta_0, \quad (19)$$

$$wage_{[compwork=1, comphome=0]} = \beta_0 + \beta_1, \quad (20)$$

$$wage_{[compwork=0, comphome=1]} = \beta_0 + \beta_2, \quad (21)$$

$$wage_{[compwork=1, comphome=1]} = \beta_0 + \beta_1 + \beta_2 + \beta_3. \quad (22)$$

Comparing the first and second equations, we obtain

$$\beta_1 = wage_{[compwork=1, comphome=0]} - wage_{[compwork=0, comphome=0]}. \quad (23)$$

Therefore, β_1 is equal to the difference in wages between those who only use computer at work and those who never use computer.

Comparing the first and third equations, we obtain

$$\beta_2 = wage_{[compwork=0, comphome=1]} - wage_{[compwork=0, comphome=0]}. \quad (24)$$

Therefore, β_2 is equal to the difference in wages between those who only use computer at home and those who never use computer.

Comparing the first and fourth equations, we obtain

$$\beta_1 + \beta_2 + \beta_3 = wage_{[compwork=1, comphome=1]} - wage_{[compwork=0, comphome=0]}. \quad (25)$$

Therefore, $\beta_1 + \beta_2 + \beta_3$ is equal to the difference in wages between those who use computer both at work and at home and those who never use computer. Note that this effect is not simply adding β_1 and β_2 . Here, β_3 is the **extra effect (also called the interaction effect)** of using computer both at work and at home (for example, one might perform more efficiently by using computer at both places and thus earn more wages/bonus. Then, β_3 will be positive in this case).

[Note: Both *compwork* and *comphome* are dummy variables taking values 1 and 0, so there is **no slope parameter** in this model, since slope parameter is only for continuous variables such as STR and GPA. In fact, here we have specified **four different levels of wages** for four types of people: those who never use computer, those who only use at work, those who only use at home, and those who use at both places. The model in this question is called a (binary variable) **interaction regression model**, which is very useful in practice when we want to study the interaction effect of certain variables.]

5. (a) STATA command: **reg ahe age, robust**; The estimated effect is 0.60.

. reg ahe age, robust

```
Linear regression               Number of obs   =    7,711
                                F(1, 7709)      =    225.70
                                Prob > F         =    0.0000
                                R-squared        =    0.0290
                                Root MSE     =    9.9919
```

	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
ahe						
age	.6049863	.0402701	15.02	0.000	.526046	.6839266
_cons	1.082275	1.167013	0.93	0.354	-1.205388	3.369938

- (b) STATA command: **reg ahe age female bachelor, robust**

With the multiple regression, the estimated effect of age on earning is 0.58. For $H_0 : \beta_1 = 0$, $t = 16.02 > 1.96$, so we can reject H_0 at the 5% level. **Also, we can reject because p -value = 0.000 < 5%, and the 95% confidence interval does not include 0.**

. reg ahe age female bachelor, robust

```
Linear regression               Number of obs   =    7,711
                                F(3, 7707)      =    555.48
                                Prob > F         =    0.0000
                                R-squared        =    0.1998
                                Root MSE     =    9.0718
```

	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
ahe						
age	.5852144	.0365302	16.02	0.000	.5136052	.6568236
female	-3.664026	.2076129	-17.65	0.000	-4.071003	-3.257048
bachelor	8.083001	.2126945	38.00	0.000	7.666062	8.499994
_cons	-.6356977	1.083075	-0.59	0.557	-2.758818	1.487423

[We also note that R^2 increased from 0.0298 in the single regression to 0.1998, but the estimate for the effect of *age* remains very similar (0.60 vs 0.58). This can be explained by looking at $\text{corr}(\text{age}, \text{female})$ and $\text{corr}(\text{age}, \text{bachelor})$:

correlate age female bachelor

We find that (1) $\text{corr}(\text{age}, \text{female}) = -0.0307$ and (2) $\text{corr}(\text{age}, \text{bachelor}) = 0.0002$. Given the small correlations, controlling for *female* and *bachelor* or not will not largely affect the estimate for *age*.]

- (c) For Bob: $\widehat{ahe} = -0.63 + 0.58 \cdot 26 - 3.66 \cdot 0 + 8.08 \cdot 0 = 14.45$. ($\widehat{ahe} = -0.64 + 0.59 \cdot 26 - 3.66 \cdot 0 + 8.08 \cdot 0 = 14.7$ if you prefer rounding. Both are fine.)

For Jane: $\widehat{ahe} = -0.63 + 0.58 \cdot 30 - 3.66 \cdot 1 + 8.08 \cdot 1 = 21.19$.

```
. margins, at(age=26 female=0 bachelor=0) at(age=30 female=1 bachelor=1)
```

	Delta-method					
	Margin	Std. Err.	t	P> t	[95% Conf. Interval]	
_at						
1	14.57988	.1928156	75.62	0.000	14.20191	14.95785
2	21.33971	.1908257	111.83	0.000	20.96564	21.71378